Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

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1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether these effects undo the innovator's dilemma found by Igami.

2 Model

2.1 State and Firm Types

The market state at each period t is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- n_o: old incumbents (have *not* adopted new technology),
- n_b: "both" incumbents (have adopted; produce with new tech),
- n_n: new entrants (using new technology),
- n_{pe}: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$.

"Both" firms are multi-product: they continue to sell old-gen (at cost c_o) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms (n_o) compete only in the old-gen market at cost c_o. New entrants (n_n) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms (n_b) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect, in that their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \text{EVT1}(0, \sigma)$ for each action a.

3. **Discrete choices** (sequential order: old $\to$ both $\to$ new $\to$ pe):

   - old: $\{\text{stay}, \text{innovate}, \text{exit}\}$
   - both/new: $\{\text{stay}, \text{exit}\}$
   - pe: $\{\text{enter}, \text{quit}\}$

   Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t + 1$.

2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms, capturing the knowledge-spillover or agglomeration channel.

2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is effectively a *single-agent problem*: earlier-mover uncertainty has been resolved, and later-mover choices enter only through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\big(\bar{V}_a(\tau, \tilde{s})/\sigma\big)}{\sum_{a'} \exp\big(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\big)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \text{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \text{CCP}_{\text{later}}]\,. \tag{5}$$

Here $\text{CCP}_{\text{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

2.5 Solution Method

The model is solved by backward induction, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state s_0, stages are solved in *reverse move order* (PE $\rightarrow$ new $\rightarrow$ both $\rightarrow$ old), exploiting the sequential structure:

   - **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. We compute CCPs for all intermediate states reachable from s_0.
   - **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
   - **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 (k_{io} old firms that chose innovate) are added to n_b before this stage.
   - **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

   No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial / Multinomial draws).

3 Calibration

We summarize the parameters we use for calibration and for simulating the data in Table 1.

4 Regional Extension

4.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$. Product-market competition stays **global**, with all active firms meeting in the same two-market Cournot game of equations (1)–(2). Agglomeration spillovers, in contrast, are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across

Table 1: Parameter values for calibration and simulation

Parameter	Description	Value
A	Demand intercept	3.0
B	Demand slope	1.0
M	Market size	1.0
c_o	Old-tech marginal cost	1.5
$c_{n,0}$	New-tech baseline cost	0.5
β	Discount factor	0.9
κ	Innovation cost	0.3
ϕ	Entry cost	0.2
σ	EVT1 scale	0.5
γ	Agglomeration (default)	0.15
ρ	Cross-market substitution	0.5

regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq^2_{oo,r}$, $\pi_{n,r} = bq^2_{nn,r}$, and $\pi_{b,r} = b(q^2_{bo,r} + q^2_{bn,r}) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers. This is a standard **local agglomeration externality**, creating a wedge between the social and private return to innovation and opening the door to regionally targeted industrial policy.

4.2 Solution with Regional States

The sequential-move structure (old $\to$ both $\to$ new $\to$ pe) carries over unchanged, and we extend the same idea one level down: **within each stage, regions move sequentially**, $r = 1 \to 2 \to 3$. When firms in region r decide, they observe the realized outcome of regions $1, \ldots, r-1$ and integrate over the (yet-to-play) strategies of regions $r+1, \ldots, R$. This is the regional analogue of Igami's cross-type sequencing: it makes every sub-stage a single-agent problem whose CCP is a *scalar* fixed point rather than a joint fixed point across regions, and it removes any coupling in the solver between regions at the same stage.

Concretely, we solve each stage in reverse region order. Region R's scalar FP is computed first for every reachable sub-state, and cached. Region $R-1$'s FP is then computed given the cached region-R continuation, and so on down to region 1. Across stages, the usual layered caches still apply: stage-4 (PE) depends only on the post-new-stage state and is reused across initial states via $\mathrm{EV}_{\text{after pe}}[s_{an}, r]$; stages 3, 2, 1 cascade through $\mathrm{EV}_{\text{after new}}$, $\mathrm{EV}_{\text{after both}}$, and $\mathrm{EV}_{\text{after old}}$, each keyed by (State, r). Because each sub-state's CCP is a function of the realized history, the per-region CCPs reported in the results below are *marginalised* over the equilibrium play of earlier-moving regions, computed by a forward traversal of the sub-stage tree.

Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about 1.8×10^4 enumerated states) and pick initial states with total firm count ≤ 6. One visible artefact of the sequential-move convention is a small first-mover asymmetry: at any symmetric s_0 with uniform γ_r, the three regions' CCPs differ by at most 10^{-3}. Region 1 commits without observing anyone, while region 3 best-responds to the realised history. The gap is well below the magnitude of every counterfactual difference reported in Section 6.

5 Estimation on Simulated Data

As a first pass at recovering the model's structural parameters from data, we simulate a firm-level dataset from the regional model under the calibration of Table 1 and then estimate (γ, κ, ϕ), treating the demand parameters, the EVT1 scale σ, the cost $c_{n,0}$, and the state cap $N_{\max}$ as known.

5.1 Simulated dataset

The simulator (`code/scripts/simulate_data.jl`) draws 500 independent markets from the 2-period regional model. Each market gets its own initial state $s_0 \sim \text{unif}\{0, 1, 2\}^{4R}$ (rejecting empty or over-capacity draws), so the panel covers a range of local cluster sizes. For every firm in every period we record region, firm type, the two Cournot quantities, and realised profit. For period 1 we additionally record the discrete action. The structural parameters used to draw the data are deliberately *not* stored in the CSV.

5.2 Two-step estimator

Because equilibrium quantities are a deterministic function of the state and the spillover vector γ, identification splits cleanly: γ is pinned by the Cournot block and (κ, ϕ) by the discrete actions, conditional on γ. We exploit this with a two-step estimator (see `notes/estimation.md` for the full writeup).

Step 1: NLS for γ. For each market we reconstruct s_0 and s_1 by counting firms of each type and region in the corresponding period. We then solve

$$\widehat{\gamma} \;=\; \arg\min_{\gamma} \sum_{m,t,r,\,\text{slot}} \left(q^{\text{obs}}_{m,t,r,\text{slot}} - \hat{q}_{m,t,r,\text{slot}}(\gamma)\right)^2, \tag{7}$$

starting from $\gamma^0 = (0, 0, 0)$ and leaving γ *unconstrained*. Because the simulated quantities are noiseless, the minimum is machine zero (SSR $= 4.88e - 14$) and $\widehat{\gamma}$ equals the truth to all reported digits; this step doubles as a sanity check that state reconstruction is correct.

Step 2: MLE for (κ, ϕ). Holding $\gamma = \widehat{\gamma}$ fixed, we maximise the period-1 action log-likelihood over (κ, ϕ). The likelihood *replays* the sequential-move order used by the solver: within each stage, regions act in order $r = 1 \to 2 \to 3$, and each firm's contribution uses

the stage CCP evaluated at the sub-state its firm actually faced (not the marginal CCP integrated over the forward traversal). Per market,

$$\begin{aligned}\ell_m(\kappa,\phi) = &\sum_{r=1}^{R}\Big[k_r^{\mathrm{so}}\log p_{\mathrm{so}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{io}}\log p_{\mathrm{io}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{eo}}\log\big(1-p_{\mathrm{so}}-p_{\mathrm{io}}\big)\Big]\\ &+\sum_{r=1}^{R}\Big[k_r^{\mathrm{sb}}\log p_{\mathrm{sb}}(\tilde{s}_r^{\mathrm{BOTH}}) + k_r^{\mathrm{eb}}\log\big(1-p_{\mathrm{sb}}\big)\Big]\\ &+\sum_{r=1}^{R}\Big[k_r^{\mathrm{sn}}\log p_{\mathrm{sn}}(\tilde{s}_r^{\mathrm{NEW}}) + k_r^{\mathrm{en}}\log\big(1-p_{\mathrm{sn}}\big)\Big]\\ &+\sum_{r=1}^{R}\Big[k_r^{\mathrm{ep}}\log p_{\mathrm{ep}}(\tilde{s}_r^{\mathrm{PE}}) + k_r^{\mathrm{op}}\log\big(1-p_{\mathrm{ep}}\big)\Big],\end{aligned} \tag{8}$$

where the $k_r^{\bullet}$ are observed action counts in region r and the $\tilde{s}_r^{\bullet}$ are the sub-states produced by sequentially updating s_0 with the realised outcomes of earlier regions and stages. The total objective is $\ell(\kappa,\phi) = \sum_m \ell_m(\kappa,\phi)$, maximised by Nelder-Mead starting from $(\kappa^0,\phi^0) = (1.0, 1.0)$. The starting point is intentionally far from the calibration, so the optimiser must traverse the parameter space. Standard errors come from the BHHH estimator $V = (\sum_m g_m g_m')^{-1}$ with central finite-difference gradients.

5.3 Results

Table 2 reports the estimates against the data-generating parameters. The two-step estimator recovers $\widehat{\gamma} = (0.1500, 0.1500, 0.1500)$ exactly (Step 1), and the action MLE delivers $\widehat{\kappa} = 0.3002$ (s.e. 0.0416, truth 0.3000) and $\widehat{\phi} = 0.1777$ (s.e. 0.0407, truth 0.2000). Both κ and ϕ lie within roughly one standard error of their true values; given 500 markets and a handful of deciders per market, this is about the precision the sample supports. The period-1 action log-likelihood at the optimum is -1851.916.

Table 2: Two-step estimation of (γ, κ, ϕ)

Parameter	Truth	Estimate	S.E.
κ	0.3000	0.3002	0.0416
ϕ	0.2000	0.1777	0.0407
γ_1	0.1500	0.1500	—
γ_2	0.1500	0.1500	—
γ_3	0.1500	0.1500	—

Notes: Estimates from 500 simulated markets. Step 1 identifies γ from Cournot quantities; Step 2 estimates (κ, ϕ) by period-1 action MLE conditional on $\widehat{\gamma}$. Standard errors (BHHH) are reported only for the MLE step.

6 Counterfactual: EU–US Alliance

Pooling regions 1 and 2's innovator counts lifts welfare by roughly +6.44 % in each allied region, leaves region 3 unchanged in welfare terms, and raises aggregate welfare by +4.31 % of baseline. Innovation rates rise by +5.16 % and +4.43 % in the allied regions and fall by $|-4.00|$ % in region 3, so the unallied region is harmed on the real-economy margin even though it breaks even on welfare. The rest of the section sets up the spillover specification, derives the welfare formula, describes the Monte Carlo evaluation, and walks through the mechanism.

6.1 Spillover specification under the alliance

Under the baseline, region r's new-technology marginal cost depends only on its own innovators, equation (6). Under the alliance of $\{1, 2\}$, regions 1 and 2 share an enlarged pool:

$$c_{n,r,t}^{\text{alliance}} = \max\left\{0,\ c_{n,0} - \gamma_r \cdot \sum_{r' \in \text{bloc}(r)} \left(n_{b,r',t} + n_{n,r',t}\right)\right\}, \qquad \text{bloc}(1) = \text{bloc}(2) = \{1,2\},\ \text{bloc}(3) = \{3\}. \tag{9}$$

The cost floor prevents the marginal cost from turning negative; under the calibration $\gamma \leq 0.15$ this constraint never binds. The implementation adds a single `blocs` field to the parameter struct; the default `(1,2,3)` recovers equation (6) bit-for-bit.

6.2 Welfare

The two-market linear demand of equations (1)–(2) is rationalised by the quasi-linear utility $U = A(Q_o + Q_n) - \frac{b}{2}(Q_o^2 + Q_n^2) - s\,Q_o\,Q_n$ with $b = B/M$ and $s = \rho B/M$, giving consumer surplus

$$\text{CS}(s) = \tfrac{b}{2}(Q_o^2 + Q_n^2) + s\,Q_o\,Q_n. \tag{10}$$

Because Cournot is global, CS is a single number; we allocate it equally across regions (equal-population assumption). Producer surplus is local: each firm's per-period profit is charged to its own region. Innovation and entry costs are paid by the firm's region in period 1. Two-period discounted welfare per region is

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\,\text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\,\text{PS}_r(s_1) - \kappa\,k_r^{\text{innov}} - \phi\,k_r^{\text{enter}}, \tag{11}$$

which satisfies $\sum_r W_r = \text{CS}_{\text{total}} + \sum_r \text{PS}_r - \sum_r(\kappa\,k_r^{\text{innov}} + \phi\,k_r^{\text{enter}})$ by construction. The runner verifies this identity numerically and finds residuals at machine precision.

6.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario. Each market draws its own initial state s_0 from the same distribution `simulate_data.jl` uses, so the counterfactual sits on the same population the model was estimated on. The same seed (20260424) drives both scenarios, and per-market we use `MersenneTwister(seed + k)`. Identical s_0 draws and

identical EVT1 shock paths under both scenarios mean each market contributes a paired observation $(W_r^{\text{base}}, W_r^{\text{alli}})$, so the variance of the welfare difference is much smaller than the variance of either level. Headline numbers stabilise to about 0.2 percentage points by $K = 1000$.

6.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 5: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 3 reports the full per-region decomposition. Innovation rises by +5.16 % in region 1 and +4.43 % in region 2, and falls by $|-4.00|$ % in region 3. Entry rates move in the same direction with smaller magnitudes.

Table 3: EU–US alliance: per-region equilibrium and welfare

	Region 1 (US)		Region 2 (EU)		Region 3 (RoW)	
Quantity	Baseline	Alliance	Baseline	Alliance	Baseline	Alliance
$P(\text{innov} \mid \text{old})$	0.3507	0.3688	0.3495	0.3650	0.3533	0.3392
$\Delta P/P_0$	+5.16%		+4.43%		-4.00%	
$P(\text{enter} \mid \text{pe})$	0.5562	0.5723	0.5801	0.5968	0.5792	0.5624
$\Delta P/P_0$	+2.91%		+2.87%		-2.91%	
PS_r	0.8500	0.9290	0.8488	0.9290	0.8800	0.8147
Costs paid	0.0905	0.0942	0.0958	0.0993	0.0958	0.0925
W_r	2.1358	2.2734	2.1293	2.2682	2.1606	2.1608
$\Delta W_r/W_{r,0}$	+6.44%		+6.52%		+0.01%	

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, uniform $\widehat{\gamma}_r = 0.1500$. Baseline blocs $(1, 2, 3)$; alliance blocs $(1, 1, 2)$. CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

6.5 Discussion

Three points are worth highlighting.

Allied regions gain in two ways. Their pooled innovator count lowers $c_{n,1}$ and $c_{n,2}$ whenever *either* region has an innovator, raising the period-2 profit of being on new tech. This pulls more firms into innovation and entry there, which raises producer surplus, and the additional new-tech output also lowers prices globally, raising the regions' share of consumer surplus. Welfare rises by +6.44 % in region 1 and +6.52 % in region 2. The pull is sharper on

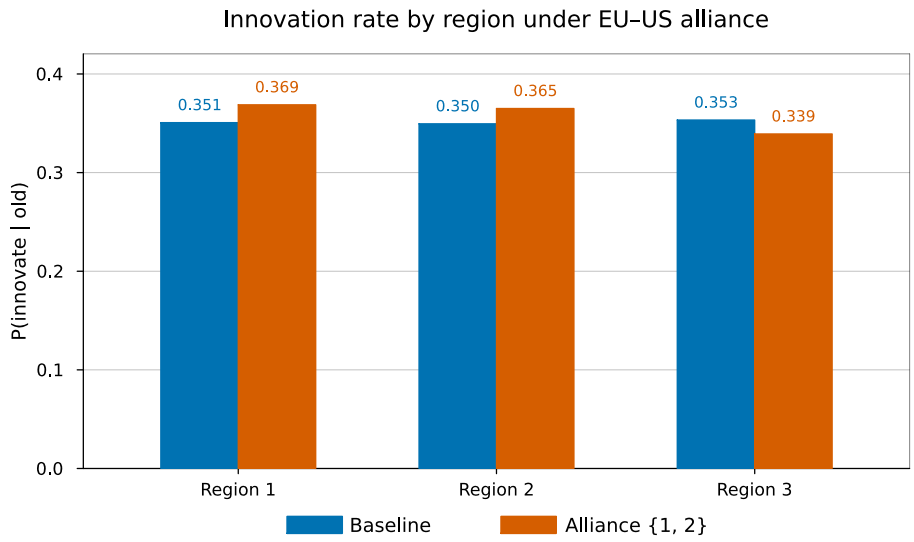

Figure 1: Innovation rate $P(\text{innov} \mid \text{old}, r)$ by region, baseline vs. alliance.

innovation than on entry. An old firm weighing "innovate" against "stay old" sees its outside option worsen as the alliance lowers c_n: cheaper new-gen output depresses the old-gen price through cross-market substitution. A potential entrant's outside option is exit at value 0, invariant to c_n, so only the upside moves and the entry response is roughly half as large.

Region 3 loses on innovation but breaks even on welfare. Tougher global Cournot competition from cheaper allied competitors drags region 3's $P(\text{innov})$ and $P(\text{enter})$ down by about 4 % and 3 % respectively, and its PS_3 falls by roughly 7 %. Its share of the global CS gain is almost the same magnitude in the opposite direction, so its net welfare moves by less than a basis point (+0.01 %). The unallied region is harmed in real-economy terms (less innovation, lower profits) but the consumer-side benefit of cheaper goods produced by the alliance compensates almost exactly.

The aggregate gain is positive and sizeable. Total welfare rises by +4.31 % of baseline. Roughly half of the gain accrues to each allied region, with a near-zero residual from region 3. Knowledge integration is a positive-sum game even before factoring in the aggregate consumer benefits.

The result depends on the calibration. Under an earlier $(\gamma, \sigma) = (0.05, 1.0)$ setting the same alliance moved welfare by under 1 % everywhere; tripling γ scales the spillover delivered by the merger, and halving σ doubles the logit slope $P(1-P)/\sigma$ against any payoff change, so the combined effect on equilibrium quantities is roughly 6×. The recalibrated values stay well below the cost-floor saturation that sets in at $\gamma \gtrsim 0.30$, so the model is in its responsive interior. Full details, including the K-stability diagnostic and the $\sigma \cdot \gamma_{\text{em}}$ welfare convention, are in `notes/merger_counterfactual.md`.

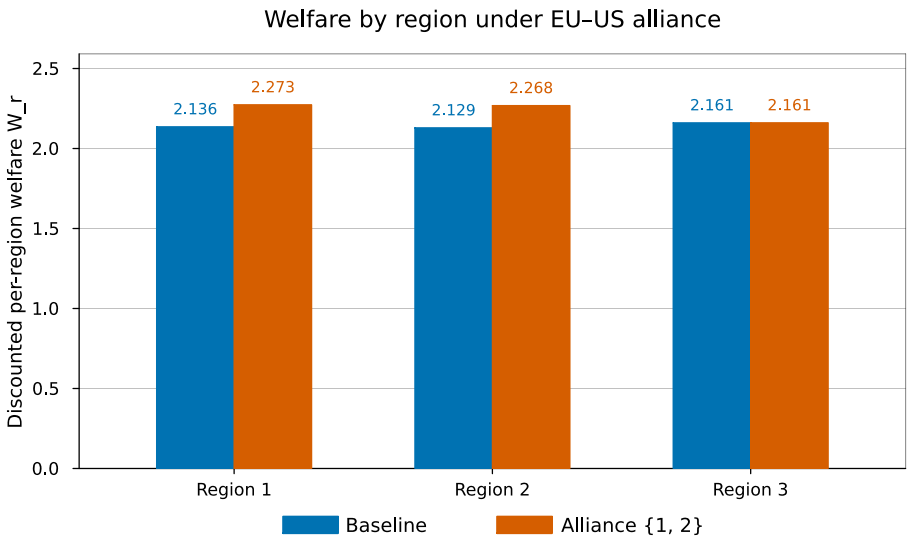

Figure 2: Per-region discounted welfare W_r baseline vs. alliance. Both allied regions gain visibly; region 3's bars are indistinguishable.

7 Counterfactual: Region-1 Innovation Subsidy

A region-1 innovation subsidy of size τ shifts the *private* cost of innovation in region 1 from $\widehat{\kappa}$ to $(\widehat{\kappa} - \tau)$ while leaving the social resource cost unchanged at $\widehat{\kappa}$. Sweeping τ over the grid $\tau/\widehat{\kappa} \in \{0, 0.10, \ldots, 0.50\}$ shows that *aggregate welfare is monotonically (weakly) decreasing in the subsidy across the entire grid.* At the largest grid point $\tau/\widehat{\kappa} = 0.50$, region 1's $P(\text{innov} \mid \text{old})$ rises by $+18.45\,\%$, but its own welfare moves by only $+0.02\,\%$, and regions 2 and 3 lose $-0.06\,\%$ and $-0.10\,\%$ respectively; total welfare changes by $-0.04\,\%$. The welfare-maximising grid point is $\tau/\widehat{\kappa} = 0.00$ ($\Delta\Sigma W = +0.00\,\%$). The rest of the section sets up the subsidy specification, derives the welfare formula under sovereign funding, describes the Monte Carlo evaluation, and walks through the mechanism.

7.1 Subsidy specification

A region-r innovation subsidy of size $\tau_r \geq 0$ changes only the *private* payoff of an old-tech firm switching to "both" technology:

$$u_r^{\text{innov}} = \pi_r^{\text{both}}(s) - (\kappa - \tau_r) + \beta\, \text{EV}_r^{\text{old}\to\text{both}}, \tag{12}$$

leaving the entry decision and the rest of the game unchanged. The implementation adds a single `subsidy` field to the parameter struct; the default `(0,0,0)` recovers the no-subsidy baseline. We sweep the one-region policy $\tau = (\tau, 0, 0)$ to isolate the question "what does targeted industrial policy in region 1 do?" from the broader question of a global R&D subsidy. The spillover specification stays at the local-only baseline of equation (6).

7.2 Welfare under sovereign funding

Each region is treated as a sovereign country: its own government funds its own subsidy from its own taxpayers, so the transfer $\tau_r \cdot k_r^{\text{innov}}$ flows from region r's taxpayers to region r's firms and cancels exactly inside region r's welfare. The *social* resource cost remains the full κ even when the firm only pays $(\kappa - \tau_r)$; the subsidy reduces the private burden, not the real labour and materials embodied in a new-technology blueprint. Two-period discounted welfare per region is therefore

$$W_r = \tfrac{1}{R}\big[\mathrm{CS}(s_0) + \beta\,\mathrm{CS}(s_1)\big] + \mathrm{PS}_r(s_0) + \beta\,\mathrm{PS}_r(s_1) - \kappa\,k_r^{\text{innov}} - \phi\,k_r^{\text{enter}}, \tag{13}$$

identical in form to equation (11): the subsidy is a within-region transfer that does not appear in W_r. With this accounting, the identity $\sum_r W_r = \mathrm{CS}_{\text{total}} + \sum_r \mathrm{PS}_r - \sum_r(\kappa\,k_r^{\text{innov}} + \phi\,k_r^{\text{enter}})$ holds exactly in the baseline and in every grid scenario. The runner verifies it numerically and finds residuals at machine precision.

7.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario. Each market draws its own initial state s_0 from the same distribution `simulate_data.jl` uses, so the counterfactual sits on the same population the model was estimated on. The same seed (20260424) drives the baseline and every grid point, with per-market `MersenneTwister(seed + k)`. Identical s_0 draws and identical EVT1 shock paths under all scenarios mean each market contributes paired observations $(W_r^{\text{base}}, W_r^{\tau})$, so the variance of the welfare difference is much smaller than the variance of either level. Headline numbers stabilise to roughly 0.05 percentage points by $K = 1000$.

7.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 5: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 4 reports the full grid sweep. Region 1's innovation rate climbs steadily from 0.3507 at $\tau = 0$ to 0.4154 at the largest grid point, while regions 2 and 3 barely move. Aggregate welfare, however, is weakly decreasing in τ across the grid, and the welfare-maximising grid point is $\tau/\widehat{\kappa} = 0.00$.

7.5 Discussion

Three points are worth highlighting.

The subsidy works on its target margin. Region 1's $P(\text{innov} \mid \text{old})$ rises from 0.3507 to 0.4154 at $\tau/\widehat{\kappa} = 0.50$, a +18.45 % lift. The mechanism is direct: a lower private cost moves the threshold draw of the EVT1 shock that triggers innovation. As a policy lever for raising

Table 4: Region-1 innovation subsidy: grid sweep over $\tau/\widehat{\kappa}$

$\tau/\widehat{\kappa}$	τ	$P_{\text{innov},1}$	$P_{\text{innov},2}$	$P_{\text{innov},3}$	ΔW_1	ΔW_2	ΔW_3	$\Delta\Sigma W$	$\Delta\Sigma W$ (%)
0.00	0.0000	0.3507	0.3495	0.3533	+0.0000	+0.0000	+0.0000	+0.0000	+0.00%
0.10	0.0300	0.3615	0.3495	0.3533	+0.0004	-0.0002	-0.0003	-0.0002	-0.00%
0.20	0.0600	0.3760	0.3495	0.3533	+0.0004	-0.0004	-0.0007	-0.0007	-0.01%
0.30	0.0901	0.3900	0.3487	0.3529	+0.0007	-0.0008	-0.0015	-0.0017	-0.03%
0.40	0.1201	0.4050	0.3487	0.3524	+0.0005	-0.0010	-0.0018	-0.0024	-0.04%
0.50	0.1501	0.4154	0.3487	0.3524	+0.0005	-0.0012	-0.0021	-0.0028	-0.04%

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across all scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, uniform $\widehat{\gamma}_r = 0.1500$. The subsidy is region-1 only: $(\tau, 0, 0)$. Welfare is computed under sovereign funding, so the transfer cancels within region 1. CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges the full $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

regional R&D activity, the subsidy works.

But aggregate welfare is weakly decreasing in τ. A subsidy is welfare-improving only if there is a positive externality on the subsidised margin that the firm fails to internalise. In our calibration the agglomeration externality $\widehat{\gamma}_r = 0.1500$ is *local*: region 1's innovation today helps only region 1's own period-2 cost, and the firm already internalises the bulk of that benefit through its expected period-2 profit. Cournot competition across regions, in contrast, makes region-1 expansion a *negative* externality on regions 2 and 3, which the planner does internalise. Both forces push the optimum toward $\tau = 0$, and the welfare-maximising grid point is exactly that: $\tau/\widehat{\kappa} = 0.00$.

The non-treated regions bear small but consistent losses. Cheaper region-1 production drags the global new-technology price down, so regions 2 and 3 see lower firm profits. The CS share they receive from cheaper goods is too small to compensate. At $\tau/\widehat{\kappa} = 0.50$, region 2 loses $-0.06\,\%$ of its baseline welfare and region 3 loses $-0.10\,\%$. These are small but signed in the same direction across all $K \in \{500, 1000, 5000\}$.

The result is calibration-dependent. Under a spillover specification that pools regions 1 and 2 (the alliance of Section 6), the planner internalises the spillover from region-1 innovation onto region 2 and the optimum would shift right; a joint EU–US innovation subsidy under blocs $(1, 1, 2)$ would be the natural comparator. With a larger $\widehat{\gamma}$ or a tighter ρ, the cross-region effects scale accordingly. Full details, including the K-stability diagnostic and the sovereign-funding accounting, are in `notes/subsidy_counterfactual.md`.

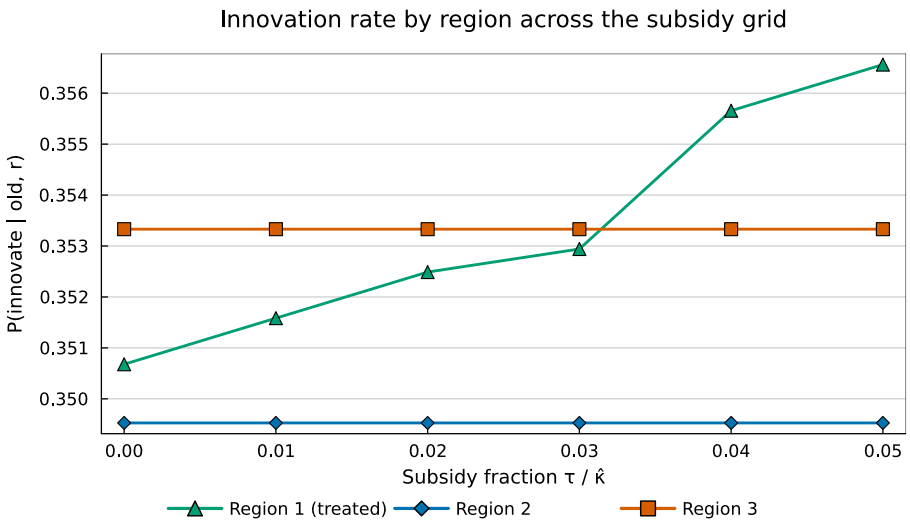

Figure 3: Innovation rate $P(\text{innov} \mid \text{old}, r)$ by region across the subsidy grid. Only region 1 (treated) responds; regions 2 and 3 are essentially flat in τ.

8 Counterfactual: Constrained Social Planner

A constrained social planner replaces firm-level decisions with logit choices that condition on social welfare differences instead of private ones. Two cuts isolate the dynamic and static channels: the *dynamic-only first best* keeps Cournot production and only internalises the spillover, while the *full first best* also imposes $P = MC$ in both periods. At the estimated calibration the full first best lifts aggregate welfare by $+8.83\,\%$ of the equilibrium baseline. Almost all of the gain is the static markup correction ($+10.68\,\%$); the dynamic-only piece is *negative* ($-1.85\,\%$). The rest of the section sets up the planner CCPs, derives the welfare formula, describes the Monte Carlo evaluation, and walks through why the dynamic-only cut falls below equilibrium even though $\widehat{\gamma}_r > 0$.

8.1 Planner specification

The planner uses the same logit form firms use, with the firm's private continuation replaced by the social one. At every sub-stage decision the choice probability is

$$\pi_a^{\text{planner}} = \frac{\exp\big((\bar{W}_a - \text{cost}(a))/\sigma\big)}{\sum_{a'} \exp\big((\bar{W}_{a'} - \text{cost}(a'))/\sigma\big)}, \tag{14}$$

where $\bar{W}_a = \mathbb{E}[\sum_r W_r \mid a_i = a]$ is the expected total social welfare conditional on the deciding firm choosing action a, integrated over peer CCPs. The planner inherits the same EVT1 shock scale σ that firms use, so this is a *constrained* planner with bounded rationality at the same noise level as the firms. The OLD-stage logit picks among {stay, innovate, exit} with $\bar{W}_{\text{innov}}$ deducting κ; PE picks {enter, quit} with $\bar{W}_{\text{enter}}$ deducting ϕ; NEW and BOTH pick

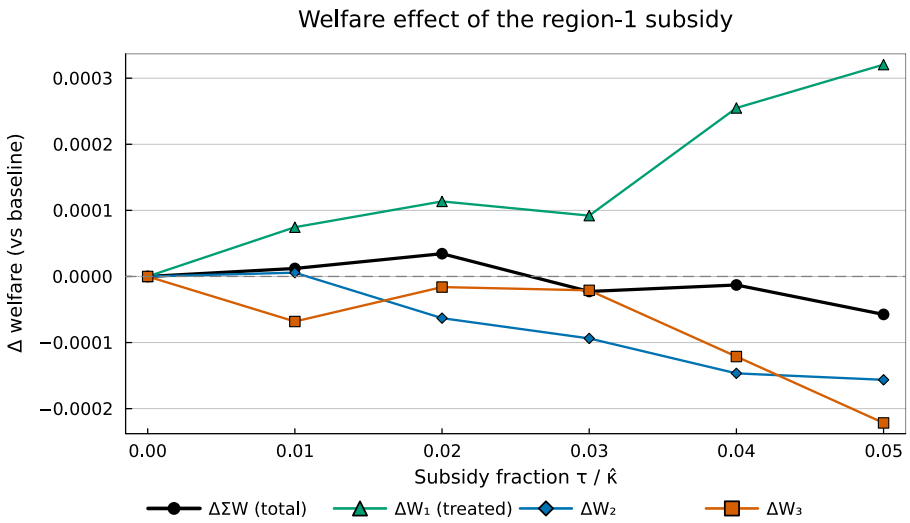

Figure 4: Per-region and total welfare change vs. baseline as a function of $\tau/\widehat{\kappa}$. Aggregate welfare (black) is weakly decreasing in τ; region 1's own welfare gain is small relative to the losses borne by regions 2 and 3.

{stay, exit} with no immediate cost on either branch.

Period-1 production happens at s_0 regardless of period-1 decisions. Period-1 social welfare is therefore independent of every stage decision and cancels in every $\bar{W}$-difference. The fixed-point machinery only needs the period-2 terminal value $V_1^{\text{soc}}(s) = \text{CS}(s) + \sum_r \text{PS}_r(s)$, computed once on the full state space. The planner solver mirrors the equilibrium solver one stage at a time (PE $\to$ new $\to$ both $\to$ old, regions $r = 1 \to 2 \to 3$ inside each stage). Full implementation in `code/src/planner.jl`.

The two cuts share the planner CCPs of equation (14) but differ in the static welfare convention used to build V_1^{soc}. Under the *dynamic-only first best*, both periods use Cournot production, isolating the spillover channel: the only difference from the equilibrium is that the planner internalises how each innovator's Δn_b lowers c_n for everyone in the same spillover bloc. Under the *full first best*, both periods use $P = MC$ pricing. With constant marginal cost, only the lowest-c_n active new-gen region produces, and old-gen output splits symmetrically across all active old-tech firms. Producer surplus is zero by construction; the entire surplus accrues to consumers.

8.2 Welfare

Two-period discounted welfare per region keeps the form of equation (11) in both planner cuts:

$$W_r = \tfrac{1}{R}\big[\text{CS}(s_0) + \beta\,\text{CS}(s_1)\big] + \text{PS}_r(s_0) + \beta\,\text{PS}_r(s_1) - \kappa\,k_r^{\text{innov}} - \phi\,k_r^{\text{enter}},$$

with the surpluses evaluated at Cournot or competitive ($P = MC$) quantities depending on the cut. Under the full first best $\mathrm{PS}_r \equiv 0$ in every region, so the identity $\sum_r W_r = \mathrm{CS}_{\mathrm{total}} + \sum_r \mathrm{PS}_r - \sum_r (\kappa\, k_r^{\mathrm{innov}} + \phi\, k_r^{\mathrm{enter}})$ collapses to $\sum_r W_r = \mathrm{CS}_{\mathrm{total}} - \sum_r (\kappa\, k_r^{\mathrm{innov}} + \phi\, k_r^{\mathrm{enter}})$. The runner verifies the identity numerically for all three scenarios; residuals are at machine precision ($\leq 3 \times 10^{-13}$).

8.3 Evaluation by Monte Carlo with common random numbers

We average $K = 5000$ independent simulations per scenario, with the same draw scheme as the merger and subsidy counterfactuals: per-market `MersenneTwister(seed + k)`, $s_0 \sim$ `random_s0(rng, p)`, seed = 20260424. Identical s_0 draws across the equilibrium, dynamic-only, and full first best scenarios mean each market contributes a triple $(W_r^{\mathrm{eq}}, W_r^{\mathrm{dyn}}, W_r^{\mathrm{fb}})$, so the welfare differences $\Delta\Sigma W$ have much smaller variance than the levels.

The planner's CCPs and continuation values are deterministic given s_0, so we do *not* resample EVT1 shocks. Sampling stays at the s_0 level only.

8.4 Calibration and results

We hold the demand parameters, β, σ, ρ, c_o, $c_{n,0}$, and $N_{\max}$ at the values of Table 1, and use the structural estimates from Section 5: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, and uniform $\widehat{\gamma}_r = 0.1500$.

Table 5 reports the per-region decomposition for the three scenarios and Table 6 reports the two welfare deltas plus the static slice. Aggregate welfare under the full first best is 6.9930 versus 6.4257 at equilibrium, a +8.83 % gain. The dynamic-only first best aggregates to 6.3065, $|-1.85|$ % *below* equilibrium. The static slice (full first best minus dynamic-only) is +10.68 %. Innovation rates fall under both planner cuts, from 0.3507 at equilibrium to 0.2984 under the dynamic-only first best to 0.2751 under the full first best (region 1 numbers; the other regions move in lockstep under uniform $\widehat{\gamma}_r$).

8.5 Discussion

Three points are worth highlighting.

The static markup correction is the larger of the two gains. At Cournot prices every active firm earns positive PS at the cost of deadweight loss. Imposing $P = MC$ turns those rents into consumer surplus and eliminates the triangle, leaving total surplus higher even though producer surplus drops to zero. Total welfare rises by +10.68 % from this channel alone, and the piece is independent of the spillover. Any model with constant-marginal-cost Cournot competition would deliver a similar gain.

The dynamic-only first best falls below equilibrium because business-stealing dominates the spillover. Two forces work against each other in the OLD-stage decision. The spillover pushes equilibrium innovation *below* the social optimum: when a firm innovates

Table 5: Constrained planner: per-region equilibrium and welfare across the three scenarios

	Region 1	Region 2	Region 3	Total/avg.
Equilibrium				
$P(\text{innov} \mid \text{old})$	0.3507	0.3495	0.3533	—
$P(\text{enter} \mid \text{pe})$	0.5562	0.5801	0.5792	—
PS_r	0.8500	0.8488	0.8800	2.5789
Costs paid	0.0905	0.0958	0.0958	0.2821
W_r	2.1358	2.1293	2.1606	6.4257
Dynamic-only first best				
$P(\text{innov} \mid \text{old})$	0.2984	0.2996	0.3039	—
$P(\text{enter} \mid \text{pe})$	0.5427	0.5547	0.5679	—
PS_r	0.8738	0.8789	0.9170	2.6697
Costs paid	0.0825	0.0868	0.0881	0.2574
W_r	2.0894	2.0902	2.1269	6.3065
Full first best ($P = MC$)				
$P(\text{innov} \mid \text{old})$	0.2751	0.2749	0.2779	—
$P(\text{enter} \mid \text{pe})$	0.5042	0.5157	0.5293	—
PS_r	0.0000	0.0000	0.0000	0.0000
Costs paid	0.0764	0.0802	0.0814	0.2380
W_r	2.3339	2.3302	2.3289	6.9930

Notes: Monte Carlo over $K = 5000$ independent markets, each drawing its own s_0 from the DGP used by `simulate_data.jl`; common random numbers across all scenarios (seed = 20260424). Calibration: $\widehat{\kappa} = 0.3002$, $\widehat{\phi} = 0.1777$, uniform $\widehat{\gamma}_r = 0.1500$. The dynamic-only first best uses planner CCPs with Cournot production in both periods; the full first best uses planner CCPs with $P = MC$ production (so $\text{PS}_r = 0$ by construction). CS allocated equally across the three regions; PS_r aggregates own-region firms' profits in both periods; costs paid charges $\kappa\, k_r^{\text{innov}} + \phi\, k_r^{\text{enter}}$ to region r.

it lowers c_n for every new-gen producer in its bloc, a benefit it does not internalise. With constant marginal cost and a fixed innovation cost κ, the marginal social value of one more new-gen producer is decreasing in the number already producing. Adding the first new-gen producer in a region kills the entire deadweight-loss triangle. Adding the fifth barely makes a dent. The firm's private gain from innovating, roughly $\pi_b - \pi_o$ in steady state, does not fall in the same way: the firm always values its own access to the lower-MC technology. The result is a Mankiw-Whinston-style excess-entry effect. Firms over-innovate from the social perspective at moderately populated states, which are typical of s_0 under `random_s0`. At our calibration the saturation force dominates the spillover, so the planner's $P(\text{innov})$ sits below the equilibrium's, 0.2984 versus 0.3507 in region 1.

Reporting both cuts disentangles two distinct policy questions. The static slice answers what moving Cournot toward $P = MC$ would deliver. The dynamic slice answers what internalising the spillover would deliver. At our calibration the static lever is

Table 6: Constrained planner: aggregate welfare deltas vs. equilibrium

	$\Delta\Sigma W$	$\Delta\Sigma W/\Sigma W_{\text{eq}}$
Dynamic-only FB − Eq.	-0.1191	-1.85%
Full FB − Eq.	+0.5673	+8.83%
Static slice (Full FB − Dyn)	+0.6865	+10.68%

Notes: Aggregate welfare differences relative to the equilibrium baseline of Table 5, in absolute units and in percent. The static slice is the gain from imposing $P = MC$ on top of the planner CCPs (full first best minus dynamic-only first best).

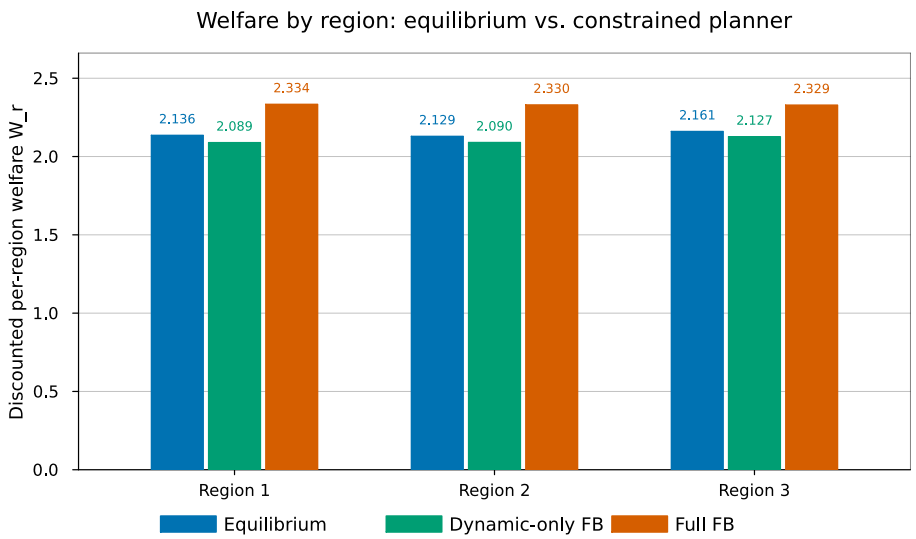

Figure 5: Per-region discounted welfare W_r under the equilibrium, the dynamic-only first best, and the full first best. The full first best gains in every region; the dynamic-only first best loses in every region.

about six times the absolute size of the dynamic lever, and the two effects are additive ($+10.68 + -1.85 = +8.83$ exactly, by construction). Either question can be studied without committing to the other; merging them into a single "first best" number would obscure that the spillover channel by itself loses welfare at this calibration while the markup channel by itself wins big.

The result depends on the calibration. With a larger $\widehat{\gamma}_r$ the spillover would be worth more at the margin and could overturn the saturation force; with a smaller ρ the static markup correction would shrink because the two product markets would be closer to independent. Full details, including the welfare-identity diagnostic, the K-stability check, and the diagnostic comparing $\beta\,\Delta V_1^{\text{soc}}$ across s_0 configurations, are in `notes/planner_counterfactual.md`.

9 Next Steps

1. **Heterogeneous absorptive capacity.** The alliance result here uses uniform $\widehat{\gamma}_r$. Allowing γ_r to differ across regions, with the bloc-pooled spec of equation (9), would let one region's higher absorptive capacity amplify the alliance's effect.

2. **Fiscal cost of policy.** The current welfare accounting prices in firms' own costs paid but does not net out the fiscal cost of any subsidy a government would need to deliver to enlarge γ_r in a target region. Adding government revenue and cost makes the targeted-subsidy and alliance scenarios directly comparable as policy levers.

3. $T > 2$ **horizon.** A genuine multi-period extension would force the value-function recursion to use the $\sigma \log \sum \exp$ ex-ante form at every internal period. The current 2-period setup avoids this because period 2 is terminal.